

### Education

- Dissertation *Tracer transport in interacting odd-diffusive systems*. Through a combination of first-principles theory, kinetic approaches, and statistical mechanics in the dilute limit, as well as field-theoretic approaches for crowded systems, I show how such symmetries, or their deliberate breaking, manifest themselves in unconventional transport phenomena of tracer particles. The particular focus of this cumulative thesis lies on odd-diffusive (also chiral) systems, i.e., systems with a broken time-reversal and/or parity symmetry. The thesis was granted with **summa cum laude** (“highest honours”). Reviewers: Prof. Ralf Metzler, Prof. Abhinav Sharma and Prof. Holger Stark
- 09/2022-12/2025 **Doctorate in theoretical physics**, *University of Potsdam*, Germany
- Master Thesis *Diffusion under the effect of Lorentz force*. Supervised by Prof. Jens-Uwe Sommer and Dr. Abhinav Sharma. Brownian particles under the effect of Lorentz force show an unexpected diffusive behaviour: Collisions can enhance the self-diffusion instead of reducing it, as ordinarily. The thesis was graded with **highest grade 1.0** and published in the Springer-Nature book-series **BestMasters**. The research was published as a *Physical Review Letter* and selected as an **Editors’ Suggestion**.
- 04/2020-05/2022 **Master of physics**, *Technical University of Dresden*, Germany, Minor: philosophy
- 07/2019-12/2019 **Semester abroad**, *Norwegian University of Science and Technology*, Trondheim, Norway, enrolled as master student
- Bachelor Thesis *Entropy production in a non-equilibrium system of hard rods in confinement*. Supervised by Prof. Jens-Uwe Sommer and Dr. Abhinav Sharma. Density Functional Theory was used in equilibrium and in dynamics to study the entropy production of interacting particles. The thesis was graded with **highest grade 1.0**
- 10/2016-01/2020 **Bachelor of physics**, *Technical University of Dresden*, Germany, Minor: philosophy
- 07/2016 **Abitur with highest grade 1.0**, *diploma from german secondary school, qualifying for university admission or matriculation*
- 2010-2016 **Secondary school**, *Max-Steenbeck-Gymnasium Cottbus*, Germany, *secondary school with extended education in mathematics, science, computer science and technology*

### Awards

- Postdoctoral fellowship Successful candidate in the “Joint Postdoctoral and PhD Exchange Fellowships by the Tel Aviv University and the University of Potsdam” (worth approx. 5.000€).
- Michelson Prize I was awarded the “[Michelson Prize](#)” of the Faculty of Science at the University of Potsdam for best doctoral thesis of the year 2025.
- IOP Impact Award The publication “Field theory of active chiral hard disks: a first-principles approach to steric interactions” in *Journal of Physics A* has been recognised for the “[impact \[it has\] achieved in such a short period of time](#)”.
- Emergent Talent Speaker At the conference “[Venice meeting on fluctuations in small complex systems VII](#)” (2024) and “[Venice meeting on fluctuations in small complex systems VIII](#)” (2026) in Venice, Italy.

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| Best Communication      | Awarded to my talk “Interactions enhance self-diffusion in odd-diffusive systems” at the conference “ <a href="#">New Trends in Nonequilibrium Statistical Mechanics</a> ” (2023) in Erice, Sicily. |
| Springer BestMasters    | Springer Nature awards <a href="#">publication</a> to the best master’s theses which have been completed at renowned Universities in Germany, Austria, and Switzerland (2022).                      |
| PRL Editors’ Suggestion | My publication “Collisions enhance self-diffusion in odd-diffusive systems” in <i>Physical Review Letters</i> was selected as an “ <a href="#">Editors’ Suggestion</a> ” (2022).                    |

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## Academic Metrics & Services

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| Publications | 10 peer-reviewed journal articles (5 first-authored, 4 as corresponding author), 1 book                                                                                                                                                                                                                                                                                                                                                                                                             |
| Citations    | 270+ citations, h-index: 7 ( <a href="#">Google Scholar</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Conferences  | 20+ invited talks, 10+ contributing talks, and 5+ contributing posters at international conferences                                                                                                                                                                                                                                                                                                                                                                                                 |
| Organization | <ul style="list-style-type: none"> <li>○ Guest-editing a <i>New Journal of Physics</i> ‘Focus On’ issue on “<a href="#">Broken symmetries and odd transport in statistical physics</a>” together with R. Metzler and A. Sharma</li> <li>○ DPG-SKM 2025 focus session “<a href="#">Broken symmetries in statistical physics: Dynamics of odd systems</a>” together with R. Metzler and A. Sharma</li> <li>○ 4 semester <a href="#">Metzler group seminar</a> (70+ international speakers)</li> </ul> |
| Supervision  | 6 Bachelor students, 4 Master students (1 finished, 2 ongoing)                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Teaching     | <ul style="list-style-type: none"> <li>○ undergraduate teaching assistant: 1 semester mathematics for physicists, 4 semesters mathematics for engineers, 4 semesters logic for philosophy,</li> <li>○ graduate teaching assistant: 1 semester theoretical mechanics for physicists</li> </ul>                                                                                                                                                                                                       |

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## Experience

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| 10/2026-03/2027 | <b>postdoctoral research assistant</b> , <i>Institut de science et d’ingénierie supramoléculaires, Université de Strasbourg, France</i> |
| 01/2026-04/2026 | <b>postdoctoral research fellowship</b> , <i>School of Chemistry, Tel Aviv University, Israel</i>                                       |
| 02/2026-09/2026 | <b>postdoctoral research assistant</b> , <i>Institute for Physics &amp; Astronomy, University of Potsdam, Germany</i>                   |
| 09/2022-01/2026 | <b>research assistant</b> , <i>Institute for Physics &amp; Astronomy, University of Potsdam, Germany</i>                                |
| 06/2022-03/2023 | <b>assistance of tax consultancy</b> , <i>Agency of Ramona Kalz, Finsterwalde, Germany</i>                                              |
| 01/2022-03/2022 | <b>research assistant</b> , <i>Leibniz-Institute for Polymer Research Dresden, Germany</i>                                              |
| 09/2021-02/2022 | <b>script-writing</b> , <i>Institute of Analysis, Technical University Dresden, Germany</i>                                             |
| 01/2020-10/2020 | <b>internship</b> , <i>Leibniz-Institute for Polymer Research Dresden, Germany</i>                                                      |

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## Selected Talks and Conferences

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| 09/2026 | <b>Invited talk</b> at the “Venice meeting on fluctuations in small complex systems VIII”     |
| 11/2025 | <b>Invited talk</b> at the “Physical Chemistry Seminar” (Tel Aviv University)                 |
| 09/2025 | <b>Invited talk</b> at the “Spintronics and Quantum Information Seminar” (AMU Poznań)         |
| 06/2025 | <b>Invited talk</b> at the “Industrial and Applied Mathematics Seminar” (U Oxford)            |
| 06/2025 | <b>Invited talk</b> in the group of Prof. M. Cates (U Cambridge)                              |
| 10/2024 | <b>Contributing talk</b> at the conference “Third Infinity 2024”(Göttingen)                   |
| 09/2024 | <b>Contributing talk</b> at the “Venice meeting on fluctuations in small complex systems VII” |
| 05/2024 | <b>Invited talk</b> at the SIAM chapter seminar (U Potsdam)                                   |

05/2024 **Invited talk** at the “Statistical Physics and Nonlinear Dynamics Seminar” (HU Berlin)  
 04/2024 **Invited talk** in the group of Prof. H. Stark (TU Berlin)  
 10/2023 **Invited talk** at the conference “New Trends in Nonequilibrium Statistical Mechanics” (Erice, IT)  
 07/2023 **Invited talk** at the seminar of DFG-TRR 146 (JGU Mainz)  
 04/2023 **Invited talk** at the “Philosophy of Physics” seminar by U. Scheffler (TU Dresden)  
 04/2023 **Invited talk** in the group of Prof. C. Sellhuber-Unkel (RKU Heidelberg)  
 05/2022 **Invited talk** in the group of Prof. J. Brader (U Fribourg)  
 04/2022 **Invited talk** in the group of Prof. H. Löwen (HHU Düsseldorf)

## Publications

ORCID: [0000-0003-3294-7365](https://orcid.org/0000-0003-3294-7365),

\* These authors contributed equally

### A Preprints

### B Journal Articles

- B1** J. Wójcik and **E. Kalz**. [The chiral random walk: A quantum-inspired framework for odd diffusion](#). *New Journal of Physics*, (in press), 2026 @open access
- B2** J. Uhlig, J. Wójcik, R. Metzler, and **E. Kalz**. [Chirality enhances diffusion in disordered environments](#). *Journal of Chemical Physics*, 165(12):124118, 2026
- B3** F. Fadedi, **E. Kalz**, R. Metzler, and A. Sharma. [A mobility based approach to transport in chiral fluids](#). *New Journal of Physics*, 28(6):063501, 2026 @open access
- B4** **E. Kalz**\*, S. Ravichandir\*, J. Birkenmeier, R. Metzler, and A. Sharma. [Reversal of tracer advection and Hall drift in an interacting chiral fluid](#). *Physical Review E*, 113(4):L042104, 2026 @Letter
- B5** P. L. Muzzeddu\*, **E. Kalz**\*, A. Gambassi, A. Sharma, and R. Metzler. [Self-diffusion anomalies of an odd tracer in soft-core media](#). *New Journal of Physics*, 27(3):033025, 2025 @open access
- B6** A. Langer, A. Sharma, R. Metzler, and **E. Kalz**. [Dance of odd-diffusive particles: A Fourier approach](#). *Physical Review Research*, 6(4):043036, 2024 @open access
- B7** **E. Kalz**, A. Sharma, and R. Metzler. [Field theory of active chiral hard disks: A first-principles approach to steric interactions](#). *Journal of Physics A: Mathematical and Theoretical*, 57(26):265002, 2024 @open access @IOP impact award 2024
- B8** **E. Kalz**, H. D. Vuijk, J.-U. Sommer, R. Metzler, and A. Sharma. [Oscillatory force autocorrelations in equilibrium odd-diffusive systems](#). *Physical Review Letters*, 132(5):057102, 2024
- B9** **E. Kalz**\*, H. D. Vuijk\*, I. Abdoli, J.-U. Sommer, H. Löwen, and A. Sharma. [Collisions enhance self-diffusion in odd-diffusive systems](#). *Physical Review Letters*, 129(9):090601, 2022 @selected as PRL Editors’ suggestion
- B10** I. Abdoli, **E. Kalz**, H. D. Vuijk, R. Wittmann, J.-U. Sommer, J. M. Brader, and A. Sharma. [Correlations in multithermostat Brownian systems with Lorentz force](#). *New Journal of Physics*, 22(9):093057, 2020 @open access

### C Books

- C1** **E. Kalz**. [Diffusion under the Effect of Lorentz Force](#). Springer Spektrum Wiesbaden, 2022